End to End Analysis on Steam Games

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https://github.com/PatrickRinn/COSC-301-Project

Introduction

We are currently living in the most prolific era of game development in history. With thousands of games launching every year, quality is not the only requirement for success, visibility plays an important part too.

Our objective is to identify some underlying mechanics of the Steam marketplace such as:

- How pricing and platforms impact a game's reach and success.
- Trends relating to a game's genre over time.
- Identifying games that we consider "Hidden Gems" or "Cult Classics" which developers can use as case studies of games which have found distinct communities.

Ultimately, the success of a game is still largely determined by its quality, but we aim to identify other variables that are indicative of success in publishing a game on Steam.

Our Pipeline

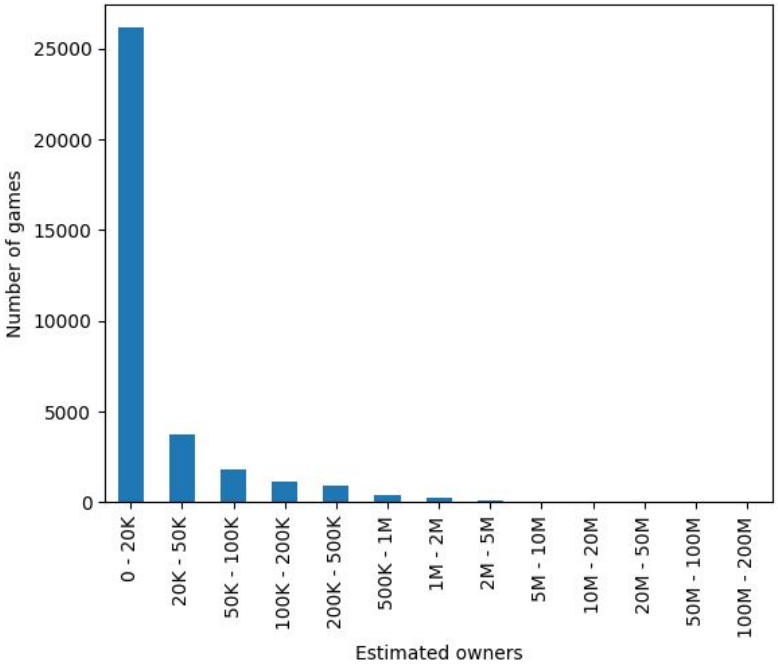
1. We created a JSON file of roughly 35,000 games with information like sales metrics and ratings. We gathered this data using publicly available code to scrape steam from this github repo: https://github.com/FronkonGames/Steam-Games-Scraper
2. Using Python, we cleaned the data and parsed it into a CSV file.
3. We then stored our cleaned data in a SQL table using sqlite3
4. We then performed our analysis on this data using Python pandas as well as Tableau.

Dataset Overview

Our dataset contains ~35,000 games, many of which have between 0-20,000 owners.

This highlights the struggle of many aspiring game developers, that many games just don't ever become popular.

~20% of games can be considered "dead", that being that they have zero public interaction (avg. playtime and reviews), and ~75% of games could be considered "ghosts", being that they have at least one review, but still show a median playtime of zero.



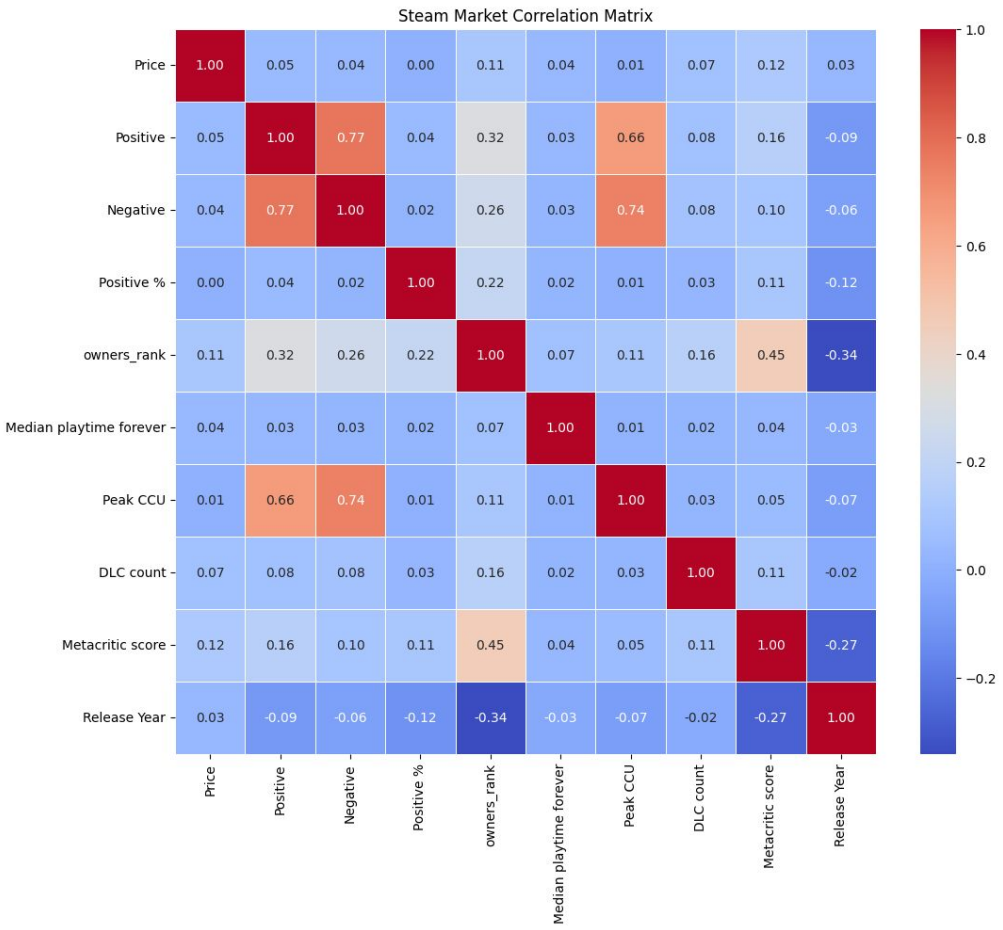
The 75% of the market made up by ghosts/dead games fail to capture even a few minutes of the player's time. As developers, our goal isn't just to avoid being "dead", but instead to reach the 25% of the market that can command player attention and are "alive".

Numeric Correlation

If we look at the correlation of numeric columns in our dataset we find:

- Volume is the biggest driver of reviews as seen by the Positive/Negative ratio (0.77)
- Popular games tend to have more negative reviews (Negative/Peak CCU)
- The average number of owners of a game decreases with newer games (Release Year/owner_rank)

This matrix ultimately highlights that Steam isn't a simple equation, and that many numerical columns are not significantly correlated with each other. This means for insights we will have to look at data aggregations and categorical columns.



Genre Analysis

This table shows the most common genres of games grouped by overall number of purchases.

From the graph we observe that in the 0 - 1,000,000 player range the most common genre is indie. However from 1,000,000 upwards, Steam becomes dominated by action games.

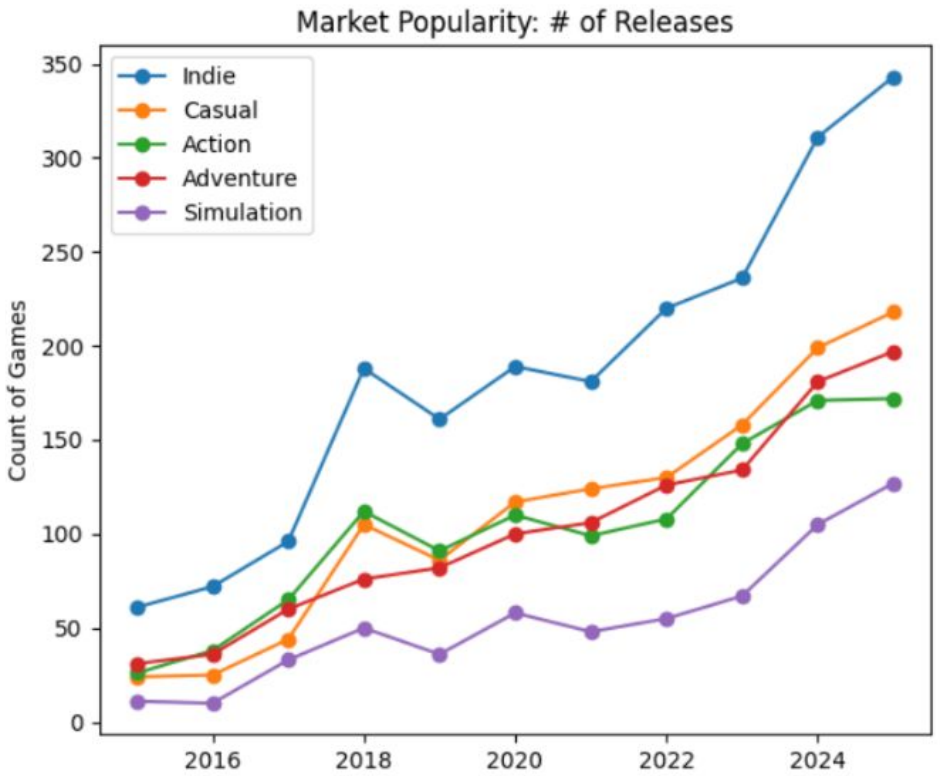
	0 - 20K	20K - 50K	50K - 100K	100K - 200K	200K - 500K	500K - 1M	1M - 2M	2M - 5M	5M - 10M	10M - 20M	20M - 50M	50M - 100M
Rank 1	Indie (73.2%)	Indie (73.8%)	Indie (70.7%)	Indie (67.3%)	Indie (64.5%)	Indie (61.6%)	Action (60.6%)	Action (62.7%)	Action (74.4%)	Adventure (68.4%)	Action (100.0%)	Action (100.0%)
Rank 2	Casual (47.0%)	Adventure (44.5%)	Adventure (44.2%)	Action (45.2%)	Action (47.8%)	Action (53.5%)	Indie (43.6%)	Indie (43.0%)	Adventure (51.2%)	Action (57.9%)	Free To Play (50.0%)	Adventure (75.0%)
Rank 3	Action (40.8%)	Action (41.7%)	Action (42.6%)	Adventure (42.8%)	Adventure (41.1%)	Adventure (41.7%)	Adventure (39.8%)	Adventure (39.4%)	Indie (27.9%)	Indie (52.6%)	RPG (40.0%)	RPG (50.0%)
Rank 4	Adventure (39.7%)	Casual (38.3%)	Casual (34.0%)	Casual (30.6%)	Strategy (27.0%)	RPG (30.9%)	Simulation (22.0%)	RPG (26.1%)	RPG (25.6%)	Simulation (47.4%)	Indie (30.0%)	Indie (25.0%)
Rank 5	Simulation (21.5%)	RPG (21.8%)	RPG (23.9%)	Strategy (25.1%)	Casual (25.5%)	Simulation (26.4%)	RPG (21.2%)	Simulation (22.5%)	Free To Play (23.3%)	RPG (42.1%)	Adventure (30.0%)	Casual (25.0%)
Rank 6	Strategy (18.7%)	Strategy (20.7%)	Strategy (23.8%)	RPG (23.7%)	RPG (24.7%)	Strategy (25.9%)	Strategy (20.3%)	Strategy (21.1%)	Simulation (23.3%)	Strategy (36.8%)	Simulation (10.0%)	Free To Play (25.0%)
Rank 7	RPG (17.1%)	Simulation (18.0%)	Simulation (21.2%)	Simulation (22.3%)	Simulation (22.6%)	Casual (23.4%)	Casual (17.4%)	Casual (14.1%)	Massively Multiplayer (18.6%)	Casual (26.3%)	Casual (0.0%)	Massively Multiplayer (25.0%)
Rank 8	Early Access (10.2%)	Early Access (7.7%)	Free To Play (11.4%)	Free To Play (11.2%)	Free To Play (11.7%)	Free To Play (14.6%)	Free To Play (14.8%)	Free To Play (12.0%)	Casual (14.0%)	Free To Play (26.3%)	Strategy (0.0%)	Strategy (0.0%)

This is most likely due to the fact that indie games are generally easier and cheaper to develop, so they flood the lower player count ranges. On the other side, less common genres like action games are generally made with higher budgets and more marketing so they can find significantly larger player bases.

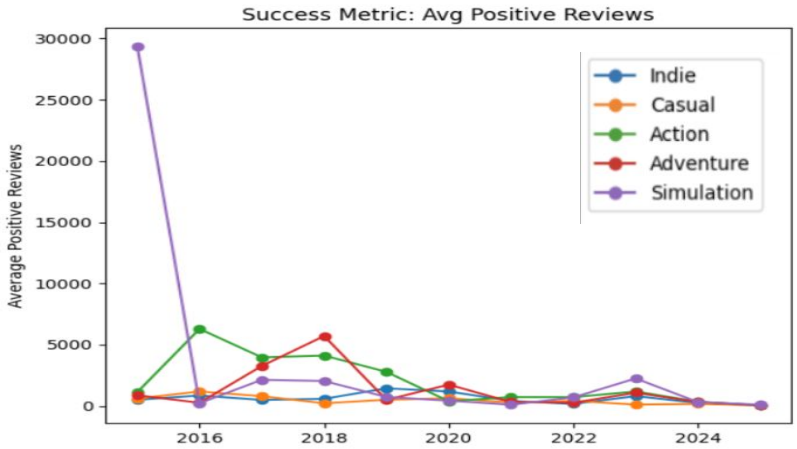
Genre: Releases by Year

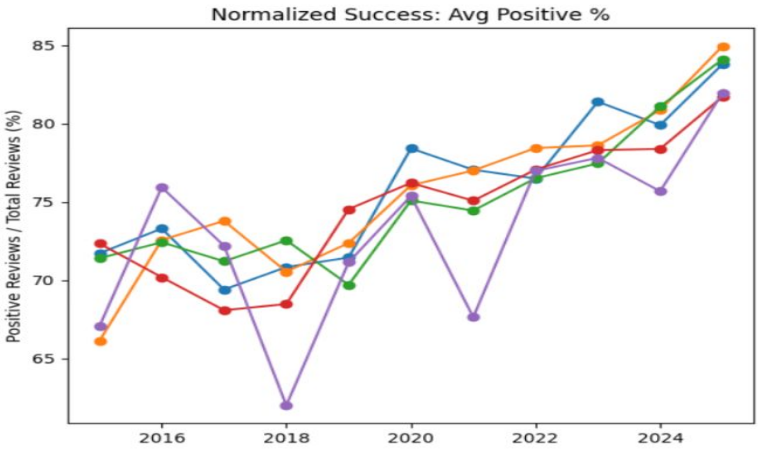
This graph shows that since 2016 the number of games released from all genres have steadily increased; however, indie games in particular have shot up substantially more than the other genres.

A possibilities for this outcome is that the tools to make indie games have become much easier for the average person to use within recent years, enabling more people to create and publish their own games.



Genre: Rating Analysis



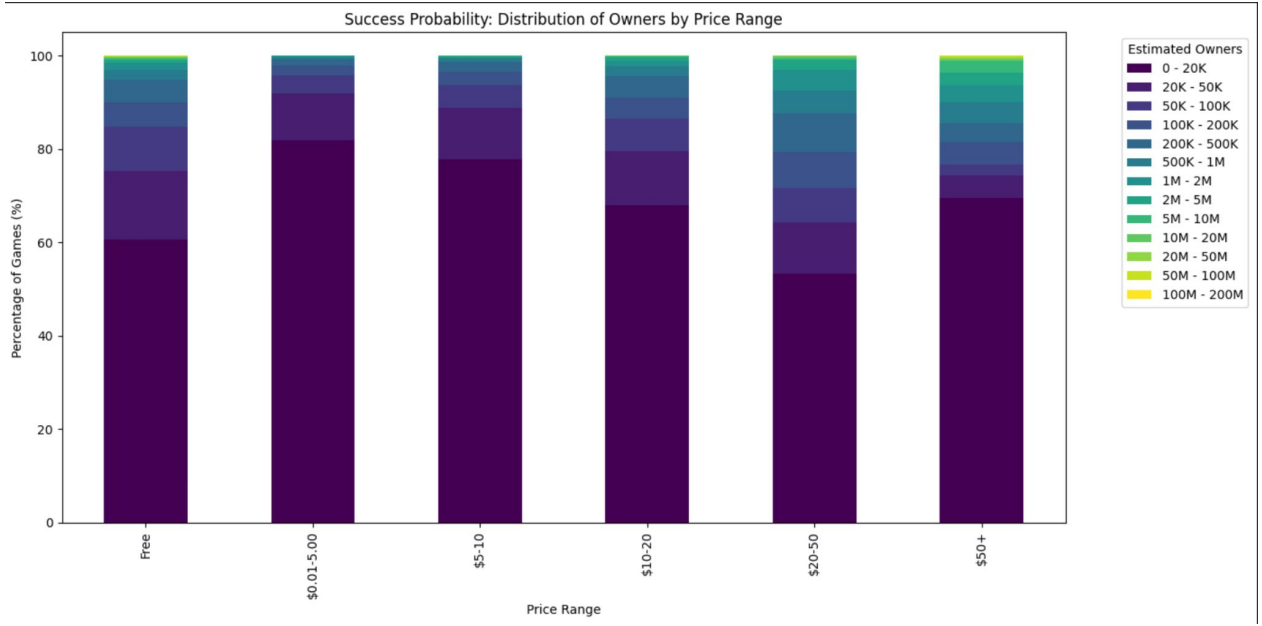


The above graphs show that since 2016, the average number of positive reviews per game has declined; however, the proportion of positive positive reviews relative to negative reviews has increased. The reason for these odd statistics can be explained by the overall decrease in reviews per game. As seen in the previous slide, each year comes with more and more games so on average there are less players playing each specific game leading to the result of less positive reviews but a higher relative percent.

Additionally, the normalized average positive rating by genre has moved consistently over time, although simulation games have had some rougher years than other genres.

Price Analysis: Success By Price Range

The Success Probability graph shows that the vast majority of games published on Steam fail to surpass the 20,000 player threshold, showing just how hard it is to create a commercially viable game.

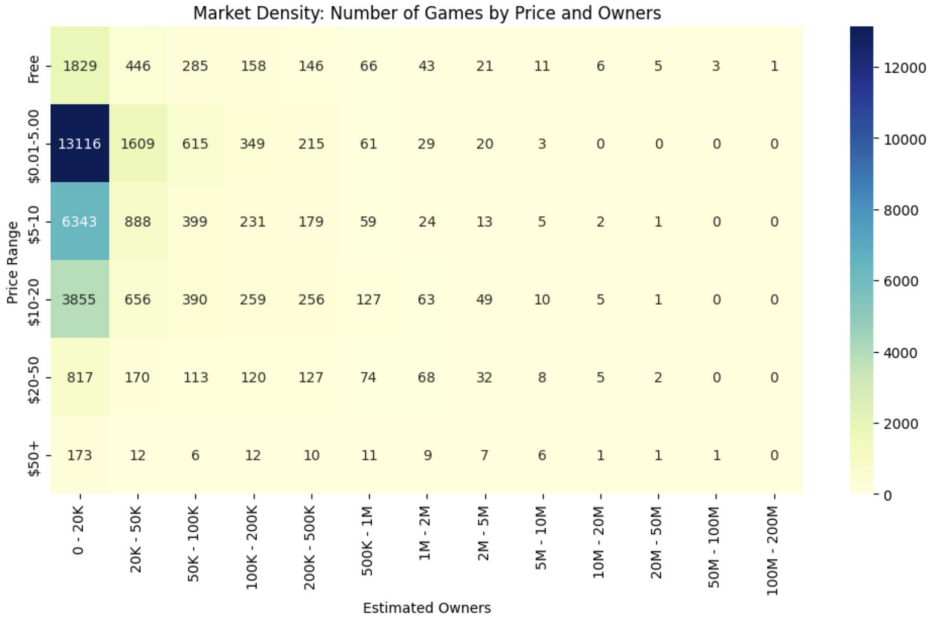


Of the few games that did find commercial success the highest success rates are found in either free games or with a 20$ plus pricetage, with games valued between that range finding minimal success.

Price Analysis: Number of Games by Price

This heat map is a continuation of the previous graph that shows the explicit amount of games in each price range and player count.

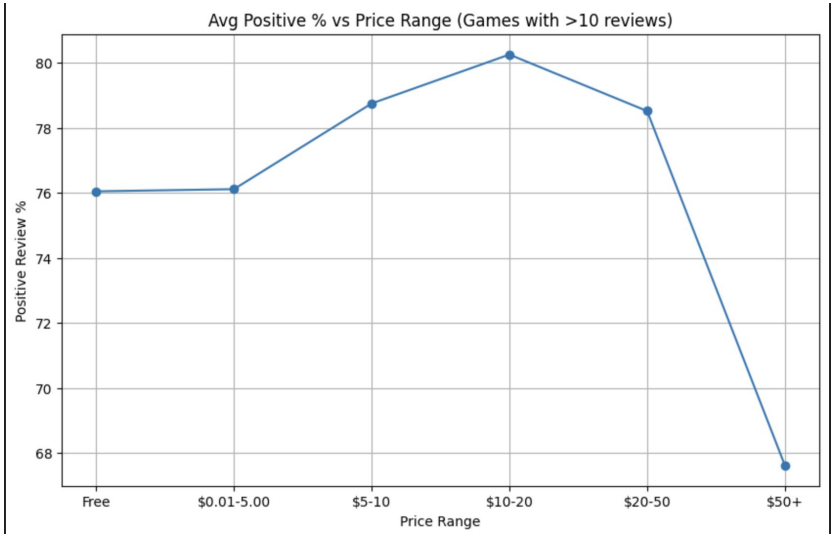
Again, like the last graph, we can see how the 0 - 20k player column dwarfs the others, showing how hard it really is to publish a successful game on steam.



Game Quality Analysis: Price Vs Reviews

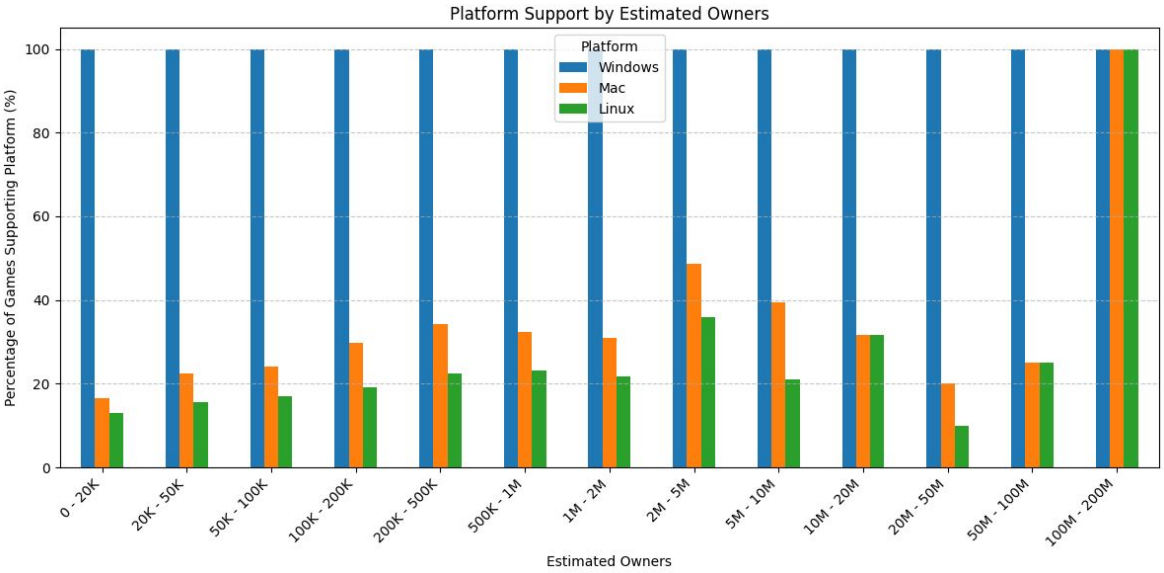
The Reviews vs. Price Range graph suggests that the best pricing strategy for maximizing positive reviews is to list a game between $10 and $20 on Steam.

This graph holds an interesting effect where above $50 the satisfaction rating drops from a high 79% to a low 68% possibly due to the fact that players no longer feel they are getting enough value compared to the cost of these games.



Platform Analysis

When comparing the estimated number of owners against the platforms the games we see that windows dominates platform support.



However, there is also a visible positive relationship between the number of owners a game has and its secondary and tertiary platform support. This suggests that making your games available on more platforms generally will increase sales.

Hidden Gems/Cult Classics

Hidden Gems:

1. Have a small but significant number of reviews (50 to 2,000) to ensure community consensus while staying "hidden".
2. Have an exceptionally high positive review ratio (>95%).
3. Have at least 30 minutes of median playtime to filter out low-effort or botted games (idlers/card farmers).
4. Are ranked by their positive review percentage and total review count.

Cult Classics:

1. Have a moderate but not massive player base (20K to 500K estimated owners).
2. Have a high positive review ratio (>80%) to ensure they are actually good games.
3. Have exceptionally high median playtime (>1,000 minutes) relative to their player count, indicating a very dedicated fanbase.
4. Exclude non-game categories like utilities or tools.

Recent Hidden Gems/Cult Classics

These games serve as the most promising inspiration for upcoming indie developers, meeting either our hidden gem criteria or cult classic criteria, and having been released in the past 5 years.

Focus Genre	Name	Release Year	Positive %	Median playtime forever	Estimated owners	Price
Action	Alien Market Simulator	2025.0	93.617021	691	0 - 20K	12.99
	Garden of Witches	2025.0	96.184739	454	20K - 50K	18.99
	Rabbit and Steel	2024.0	97.279302	411	200K - 500K	14.99
Indie	Fantasy Tavern Sextet -Vol.3 Postlude Days-	2021.0	91.428571	107448	0 - 20K	12.99
	Gentoo Rescue	2025.0	100.000000	1673	0 - 20K	19.99
	Ninja Village	2022.0	94.036697	2334	0 - 20K	8.99
Adventure	Fantasy Tavern Sextet -Vol.3 Postlude Days-	2021.0	91.428571	107448	0 - 20K	12.99
	Sankaku Renai: Love Triangle Trouble	2019.0	94.800000	5632	50K - 100K	29.99
	Tsuki ni Yorisou Otome no Sahou 2	2025.0	96.356275	3233	0 - 20K	39.99
RPG	Ninja Village	2022.0	94.036697	2334	0 - 20K	8.99
	Fuga: Melodies of Steel 3	2025.0	97.727273	1710	0 - 20K	39.99
	Esports Godfather	2024.0	91.976464	2501	200K - 500K	13.99
Strategy	Close Combat 2: A Bridge Too Far	2024.0	93.750000	3656	0 - 20K	3.59
	Gentoo Rescue	2025.0	100.000000	1673	0 - 20K	19.99
	Ninja Village	2022.0	94.036697	2334	0 - 20K	8.99

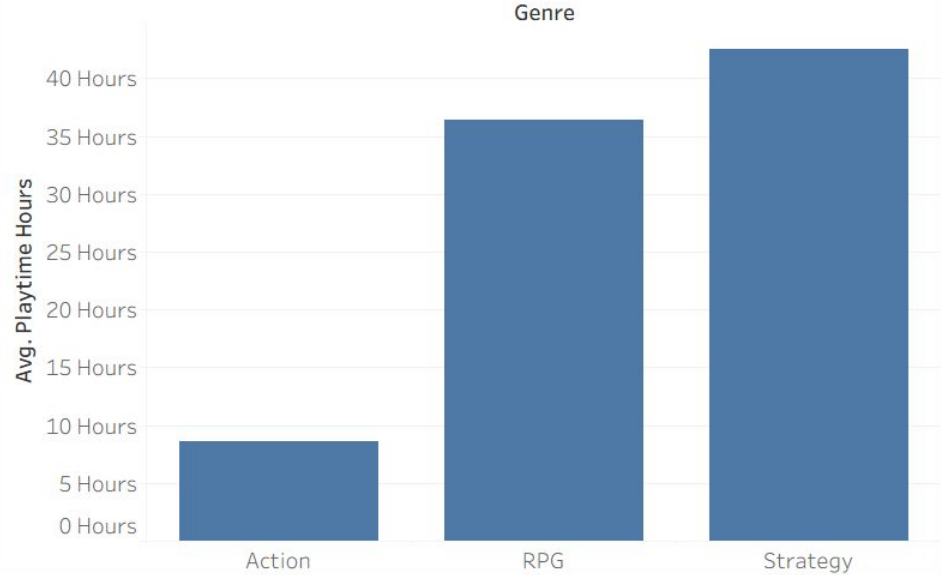
Our Findings (Tips for Developers)

1. Standing out on Steam is becoming increasingly difficult. If you are looking to release an indie game, you may have to find a novel idea as each year many more games get released and the average game gets fewer reviews
2. If your goal is commercial success with large amounts of players, action games statistically perform the best.
3. Publishing your game on platforms other than Windows may help increase the number of players your game will attract.
4. As an Indie developer, be prepared for many of your games to fail as the vast majority of games on steam never break the 20,000 player threshold
5. Pricing tips:
    a. Free: statistically more likely to have a large player base with middle of the road reviews.
    b. $10 -$20: Best for maximizing positive reviews but less likely to have a sizable playerbase.
    c. $50+ Better chance to gather a large player base but a higher risk of lower player satisfaction.

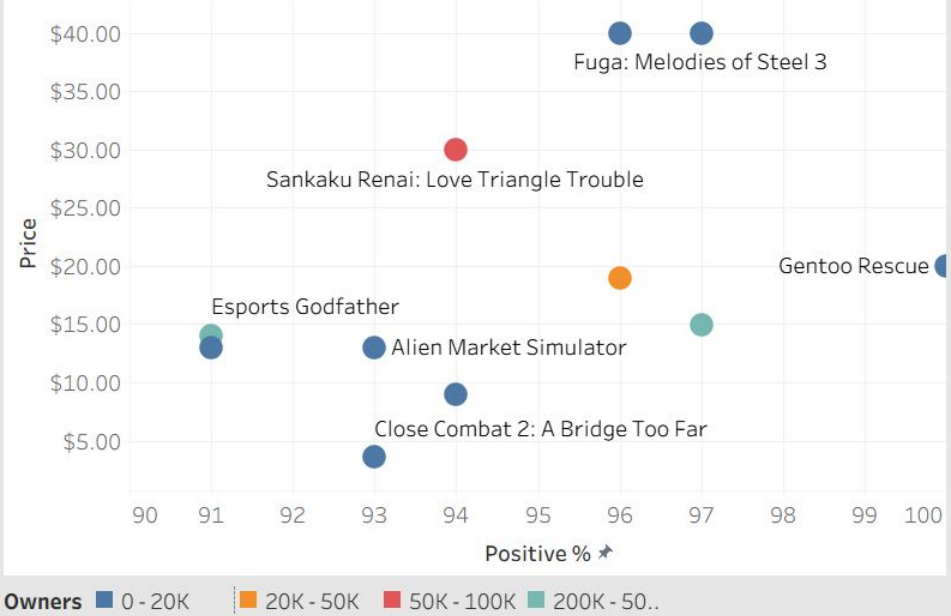
Developer Recommendations (Based on Hidden-Gem/Cult-Classic Criteria)

Expected Positive Reviews	Recommended Price	Expected Average Playtime
94.73%	$17.83 ($15-20)	29 Hours

Playtime By Genre



Opinion By Price By Owner Count



Based on our analysis, we recommend making an roguelike-action game with RPG elements. Pricing should optimally be around $15-20, reviews should be about 94% positive, and average playtime will be around 29 hours.

Our Findings (Tips For Gamers)

1. Hidden gems could be found anywhere due to the large amount of games that never found large audiences
2. A higher price does not always indicate a better game. As the price increases past the $20 mark the percentage of positive reviews tends to shrink.
3. If you are looking for games with large communities and playerbases, free and premium ($50+) games are where you should look

Limitations and Next Steps

Limitations:

Due to time and resource constraints, we were only able to gather data on 25% of the available Steam games. The process of gathering the data is very time consuming, taking about 2.73 seconds per game. Therefore, to gather all available data it would take about 125 hours of scraping, which was outside of our capabilities for this project. Another limitation are the external factors that drive a game's success that we can't capture from Steam marketplace data, such as marketing tactics, and influencer coverage.

Next Steps:

In the future we can look into:

- Collecting the rest of the games for our dataset.
- NLP analysis of reviews: Find not only games that players liked, but the things that they liked about them.
- Game discovery pipelines: How do games often find their initial surge of popularity?
- Time-series tracking: Determine if updates, sales, or seasonal events affected the copies of games sold.